Algorithmic Detection of Financial Fake News

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Abstract

Financial misinformation can distort stock market prices by generating artificial volatility, influencing investor behavior, and challenging the assumptions underlying the Efficient Market Hypothesis. This research focuses on creating a model that can detect fake news articles by recognizing them from textual features and social media interactions. Using the FakeNewsNet dataset, financial news records were filtered by searching for some words in the news articles —which are provided in the methodology section—, then by making feature engineering more features were added and some features were deleted. Two different model configurations were primarily tested. These models were based on either SVM or KNN. The SVM-based model was more accurate than the KNN-based model, performing a final accuracy of 93.88% and an F1 score of 96.05%. This model was saved for practical use in the GitHub repository (best_fake_news_model.pkl). These findings demonstrate that social media interactions, textual analysis, and the analysis of news article titles and word counts can be used to detect financial misinformation in real-world contexts, and helps to reduce market manipulation and promote a fairer investment environment.

Keywords: financial misinformation, fake news detection, SVM, KNN, BERT, spread amount, social media.

Introduction

In finance, market information quickly and directly affects the prices (Titan, 2015). The Efficient Market Hypothesis (EMH) explains that every movement in the price of any trading asset is based on information (Titan, 2015). This means that every piece of information can influence market movements, and every market movement is driven by at least one source of information. This theory explains the importance of the information because it has a direct impact on the price changes of a stock. Fake news manipulates people and controls their psychology which controls the supply-demand relation in the market (Titan, 2015). However, with the popularity of social media increasing day by day, misinformation is becoming a greater problem. Information pollution is shaping the market, creating an artificial volatility. This hypothesis is explained as: "Returning to Fama (1969 and 1970), the weak form of EMH was presented as the state of fact in which the current prices of financial assets incorporate, at any moment, all the existing historical financial information." (Titan, 2015, p. 443).

News is significant because it can directly manipulate large numbers of people, which can result in shaping the financial markets in an unfair way by creating artificial volatility. Mitigating the spread of misinformation is imperative, as its impact extends far beyond financial markets. Left unchecked, it can directly jeopardize individuals' livelihoods by causing unrecoverable financial losses. With the rise of Artificial Intelligence (AI), especially Large Language Models (LLMs), more misinformation is created by people, so this problem has become more pressing (Shu et al., 2017). Indeed, fake news is not only a tool for the spread of misinformation; it is also a type of algorithmic manipulation (Shu et al., 2017). Algorithmic manipulation poses a significant threat to modern society because it concentrates the power to influence individual choices, behavior, and access to information within the hands of a single entity or a small group. (Titan, 2015).

This issue primarily affects individuals, particularly those attempting to preserve the value of their investments or savings, as misleading information can easily trigger panic and influence their

financial decisions. Normally they do not have any impact on the market trend individually. However, if manipulated individuals accumulate, they could even change the trend in the market. This artificially generated volatility also undermines confidence and trust in the market. From an ethical perspective, the main goal is to provide everyone an equal chance to invest their money in a more trustworthy market. By providing individuals with the knowledge and tools to recognize and prevent market manipulation, the goal is to reduce unfair advantages gained by those who exploit others through deceptive practices. The ultimate objective of this research is to surpass the simple algorithms like classification, and provide an equitable investing environment for individuals by impeding misinformation.

Finance is a challenging sector for AI systems to analyze and learn from due to the high level of noise and uncertainty introduced by constantly changing news information. (Shu et al., 2017). This situation has proven that the effect of the news on the market is indisputable. Since everyone who knows financial markets knows the significance of the information, they also know that they can have control over it by employing AI tools, the internet facilitating spread of information, and exploiting the fear of individual investors (Shu et al., 2017).

There is a large volume of news content being spread around the internet, conventional media, or social media. It is impossible for humans to check the context of it and analyze the features of every news article or news program that is being published. Therefore, a dedicated tool was developed to address this objective. AI is a clear match for solving this problem since it can perform a large number of calculations in seconds, finish tasks significantly faster, and recognize patterns that are too complex for the human brain (Shu et al., 2017).

Literature Review

When training financial forecasting models, one of the most crucial aspects is the accuracy of the data given to the system (Alghamdi et al., 2024). According to EMH, the investors cannot win against the

market all the time and create enormous amounts of profits. This situation is given in the literature as: “The term ‘efficiency’ denotes the fact that investors have no opportunity of obtaining abnormal profits from capital market transactions as compared to other investors, they cannot beat the market.” (Titan, 2015, p. 442). The information is not always correct in the real world since there could be some misinformation; in addition, fake news is seriously polluting the information flow in the market. Incorrect information causes the AI models to analyze the market wrong and make them create false predictions. To perform more accurate and reliable financial analysis, it is essential to detect and filter out fake news using advanced AI models (Rusli & Yaakub, 2025).

One of the most important models for detecting fake news is Natural Language Processing (NLP), which analyzes the text provided as input to determine whether the information is likely to be genuine or misleading (Alghamdi et al., 2024). Another prominent model used in this area is exBAKE model, which is a BERT based system. The exBAKE model not only inspects the words in the news one by one, but it also analyzes the relation between the title of the news and the text of the news thoroughly. The people who produce fake news generally use exaggerated and unrelated titles to attract the reader's attention. exBAKE checks if there is any inconsistency to determine if the news article is fake or not. In the sources this model's structure is explained as: “In this paper, we propose BAKE, an automatic fake news detection model that improves upon BERT by mitigating the data imbalance hindrance. Furthermore, we also propose a model that incorporates extra unlabeled news corpora into BAKE, which we term exBAKE.” (Jwa et al., 2019, p. 2). The data imbalance that is being referenced in this section is a consequential technical problem, because in the real world the number of fake news is considerably lower than the number of the real news. This might cause a problem since the AI model can have some difficulties finding the patterns in a low number of fake news samples. The exBAKE model, in order to not face this problem, uses a special mathematical method named weighted cross entropy. By using this method, it can accurately classify the fake news, which are low in number, accurately (Jwa et al., 2019).

However, relying solely on the text and title of a news article is not sufficient to achieve accurate fake news detection in all cases. Fake news can imitate real news so well that classical basic text analysis algorithms or applications cannot classify these two different news articles (Shu et al., 2017). To overcome this limitation, AI analyzes not only the content of a news article but also its social context, identifying correlations and patterns that ameliorate the accuracy and reliability of fake news detection. This is the part why TriFN is created. The TriFN model is analyzing the relations of data about the spread of the news article at the same time. These correlations involve the person who published the news, the news article itself, and its interactions (such as likes or upvotes). The background of the publisher of the news article and the people's attitudes that are interacting with this news give strong signals to the AI about the accuracy of the news. The TriFN, which uses the data about the social context of the news article, has a lot higher accuracy in its predictions than the conventional methods (Shu et al., 2019).

In recent research, there are some models that are Graph-Attention-Network-based that have been trained (Chang et al., 2024). However, the projects that were already in the literature (exBAKE, TriFN, DIHAN) mostly focus on general social media topics and do not analyze the spread of the news article directly (Rusli & Yaakub, 2025). Furthermore, there is a gap in the literature for more practical and basic models that are connecting the text features with the social media interaction amounts (Rusli & Yaakub, 2025).

Identifying the fake news with the artificial intelligence models is a process. First, the consistency between the news article's title and its content should be evaluated. In addition, technical issues such as data imbalance should be addressed using ExBAKE-like BERT-based models. On the second step, the spread amount of the news and the relation between the publisher and the audience should be analyzed by using models like TriFN, which analyzes the social context of the news. When there is a large amount of information pollution, advanced artificial intelligence filters have become one of the most critical tools (Alghamdi et al., 2024).

Ultimately, the existing models like the exBAKE and TriFN are not focusing on financial news specifically, which can be important for individual investors, and can cause some financial problems. There is a gap in the literature about the trust analysis of financial fake news. As mentioned, the exBAKE model is focusing on textual analysis and the TriFN model is recognizing the patterns using the data of publisher trust to address the truth state of the news article. These models were missing the effect of social media interactions on recognizing fake news. So, there was a gap in the literature of the analysis of social media spread of news and financial focus.

Research Question

Can the truthfulness of financial news published on news websites be predicted using the extent of its spread and the context of the news article? This research question was created based on the gap in the literature. The aim is to find a correlation between how people interact with the news article in social media and if it is accurate or not. The words “amount of spread” in the question represent how much the news article was shared across Twitter (X). The usage of the data of the amount of spread is the key difference that this research makes. Furthermore, the category was specified to “financial news” to fill the finance gap in the literature.

Method

Repository Link: https://github.com/King-Middle/Pioneer

Google Colab Notebook: Available in the GitHub Repository

Raw and Final Dataset: Available in the GitHub Repository (named final_FakeNewsNet.csv)

A dataset that was found on Kaggle was the main dataset of this research and text or the word count of the news article were added as features by feature engineering. The dataset is named final_FakeNewsNet.csv in the provided GitHub repository. This is a dataset that contains some fake news

and real news with some features like the interaction amount or the title of the news article. The data are collected from various domains which make the examples unique so it is a great source for AI to be trained on. Every unit on this dataset represents a different news article that was published in the past on the internet with its important features.

- Important Variables:
    - Title
    - Article_text
    - Word_count
    - Tweet_num
    - Real
- Dataset Size: from 22866 rows to 489 rows
- The data is public, it is not updated regularly, secondary collection

The dataset is well-fit for analyzing the interaction between the news' popularity, user agreement and precision, as it obviously includes these crucial features. In addition, the organized structure makes it easier preprocessing the data and model integration directly. Since contextual data is not present as a diverse feature, it is obtained from the news URLs (news_url) to supplement the model with contextual information. Only the news articles that are about finance were filtered from this dataset so that it was fed with more relevant data for its intention. Additional features were added making it easier for the model to learn the correlations.

For the feature engineering part of the project, the financial news was first filtered. They were filtered by detecting some finance related specific words in the titles of these news articles, also supported by the search of the finance news publishing domain names: bloomberg.com, cnbc.com, wsj.com, reuters.com, ft.com, marketwatch.com, finance.yahoo.com, fool.com, investopedia.com, barrons.com, businessinsider.com, forbes.com, seekingalpha.com, thestreet.com, economist.com, fortune.com, money.cnn.com, kiplinger.com, bankrate.com, cq.com, cbo.gov, politifact.com, bls.gov,

commonwealthfund.org, heritage.org, taxpolicycenter.org, celebritynetworth.com, gobankingrates.com, thebalancecareers.com, networthwikibio.com, finapp.co.in, thecinemaholic.com, cbsnews.com, nbcnews.com.

The financial words that were used are: stock, stocks, stock market, stock exchange, stock price, share price, market cap, wall street, nasdaq, dow jones, s&p 500, bull market, bear market, market crash, federal reserve, the fed, interest rate, quantitative easing, monetary policy, gdp, inflation, recession, economic growth, economic policy, economic outlook, economic forecast, fiscal, unemployment rate, jobs report, payroll, labor market, consumer spending, retail sales, consumer confidence, manufacturing, industrial production, hedge fund, mutual fund, ipo, initial public offering, merger, acquisition, earnings report, quarterly earnings, revenue growth, profit margin, net income, shareholder, shareholders, trade war, tariff, tariffs, trade deficit, trade deal, economic sanction, trade sanction, bitcoin, cryptocurrency, crypto, blockchain, treasury, bond market, stimulus package, bailout, federal budget, spending bill, appropriations, debt ceiling, sequestration, budget deficit, national debt, government shutdown, outlays, receipts and outlays, tax reform, tax cut, tax plan, tax bill, tax code, tax evasion, tax fraud, corporate tax, capital gains, estate tax, tax foundation, healthcare bill, health care bill, obamacare, affordable care, medicaid, medicare, insurance premium, health insurance, housing market, mortgage rate, real estate market, subprime, foreclosure, oil price, crude oil, commodity, financial regulation, dodd-frank, sec investigation, insider trading, financial crisis, minimum wage, wage growth, net worth, in debt, bankruptcy, bankrupt, billionaire, millionaire, financial disclosure, loan, loans, lending, credit, budget and economic.

The news that did not contain these words or was not published by the financial news publishing domain was deleted from the dataset since it was out of the focus of the research question. Two different features were added, and two redundant features were deleted for the model to have a higher accuracy. The text of the news article was one of the features that were added to be able to make textual analysis

and feed the model with it so it can have more data to analyze and find a pattern in which is also related with the market. The URL of the news articles was used to be able to get this data, which was another feature in the proposed dataset before the feature engineering process.

The other feature that was added and used is the word count of this text feature that was previously added because the word count of a text can help the model to be able to make deeper textual analysis. The two features that were removed from the dataset were URL and domain of the news article since these were two features that could be used to get more data but would not be sufficient for training a model directly. After this process of adding and removing data, all remaining records with missing values from the dataset were deleted. At the end of these changes the dataset shrunk from 22866 records to only 489 records because the first version of the dataset did not only contain the financial news.

To feed the model appropriately with the textual data and numerical features a two-step attribute process has been used. In the structure of the model BERT was used to analyze the title and the content of the news article and 384-dimensional text has been converted into vector files because in literature to find the meaningful relations in texts BERT based vector models are used with the demonstrated effectiveness (Jwa et al., 2019). The numerical variables (tweet_num, word_count, and char_count) are scaled by the StandardScaler and their Z-Score is standardized. The obtained textual vectors and standardized numerical attributes were gathered for creating a hybrid attribute dataset. The dataset was distributed as 80% training and 20% testing to be able to have a balanced model with the right amount of ratio in each class (train-test split of 0.2).

In the classification phase, the Support Vector Machines (SVM) and K-Nearest Neighbor (KNN) were chosen for the highest accuracy with a simple model. Since the data imbalance between the real and fake data was one of the critical problems in the literature (Jwa et al., 2019), the SVM model was fed with balanced class weights. This method hinders the model to ignore the fake news by increasing the loss

penalties for the fake news which are the minority. SVM's ability to make more complex operations with the numerical data and the data it gets from the BERT model outperforms the simple algorithm of the KNN's simple strategy of making classifications based on the distance of the data. For finding the better model among these two trials two different metrics were used. Accuracy was used for having the simplest metric for understanding how accurate the model is, and F1 score was checked since it is both reflecting the recall and precision of the model which are important metrics on determining the sensitivity of the model.

Since only focusing on the accuracy scores of the models could lead to misinterpretation in unbalanced datasets, F1 Score was determined as the main assessing criterion. F1 Score is the harmonic average of the precision and recall scores of the model. By analyzing the confusion matrix and the ratio of the False positives and false negatives, models' wrong classification tendencies were confirmed.

Results

When the model was finalized, two different versions with different model algorithms (SVM, KNN) were tested again to record the final performance scores of these models. SVM scored 93.88% in accuracy and 96.05% in F1 score; KNN scored 92.86% in accuracy and 95.60% in F1 score. The test results of the created models are analyzed and given in Table 1.

Table 1

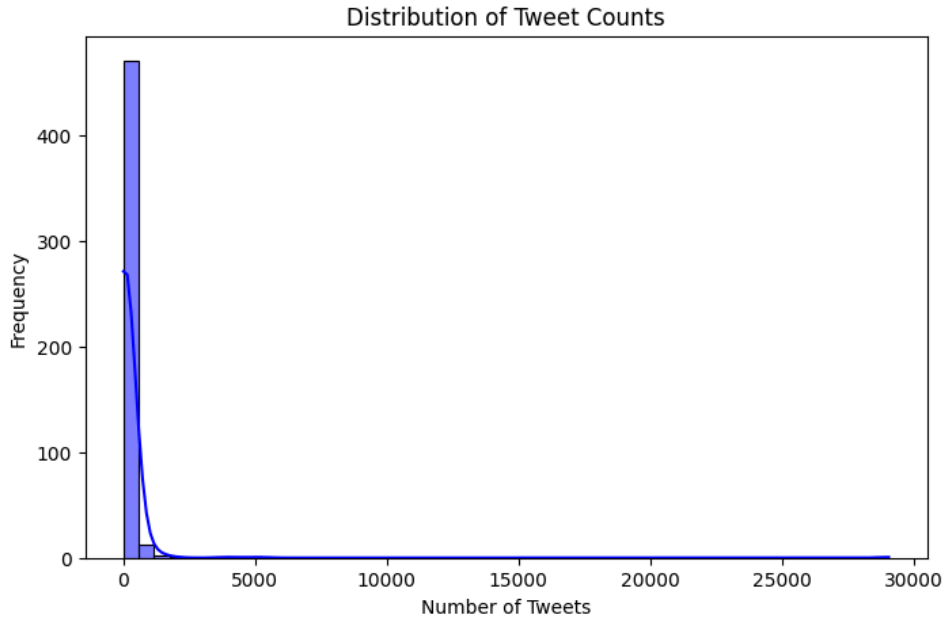
Performance Comparison of Model Algorithms

Model Algorithm	Accuracy	F1-Score	Analysis
SVM	93.88%	96.05%	Optimum model
KNN	92.86%	95.60%	Tested model

In Table 1, the SVM classifier performed better in both of the metrics than the KNN with a score of 93.88% as accuracy and 96.05% as F1 score. SVM performed better with the help of its more complex algorithms than the KNN because while SVM is an eager learner, KNN performs as a lazy learner. KNN's distance metrics are affected by the data imbalance. The details of the dataset, distributions of data and the final model were extracted by creating some graphs reflecting these.

Figure 1

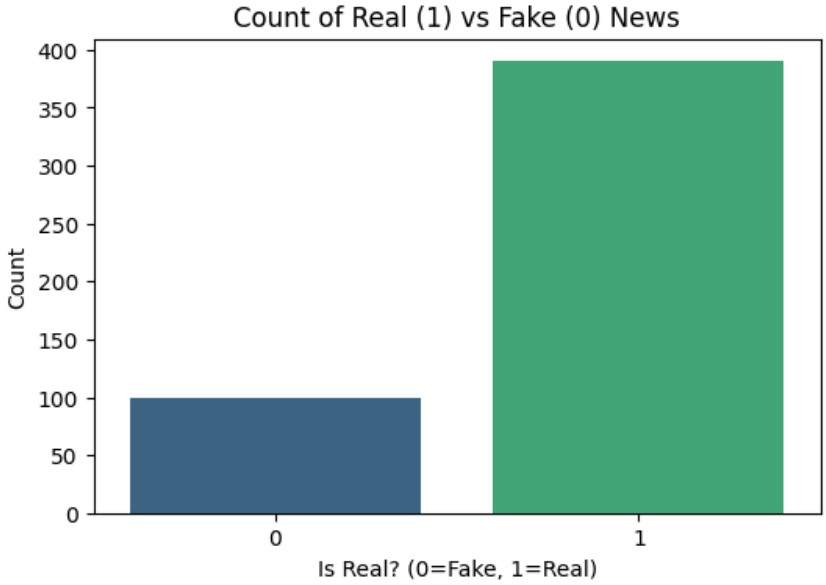
Distribution of Tweet Counts



Note. This is the distribution of the social media interactions in the dataset

Figure 2

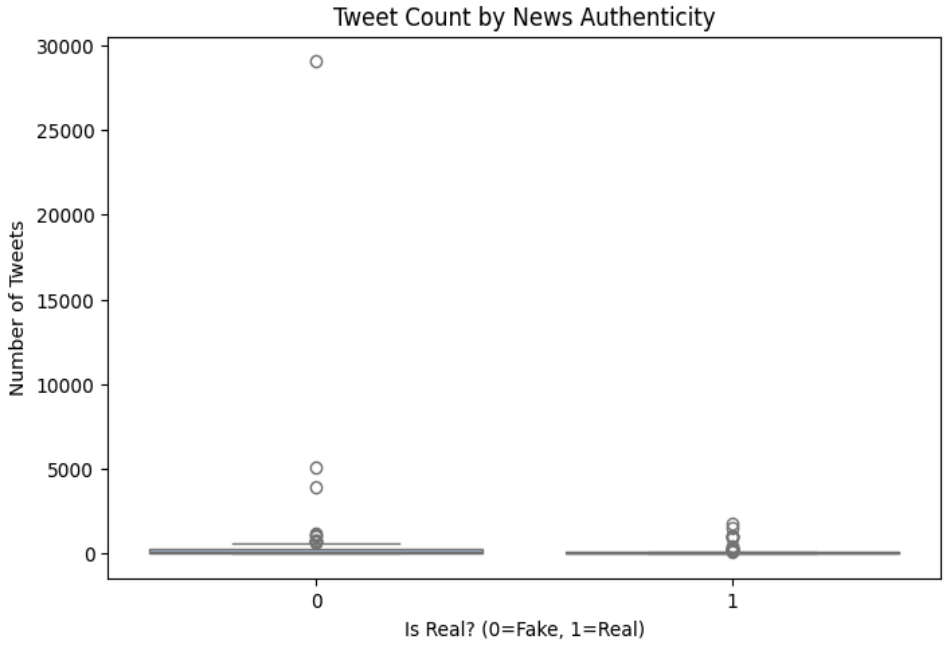
Count of Real (1) vs Fake (0) News



Note. This is a graph of the distribution of the fake news in the dataset according to the classification of real or fake.

Figure 3

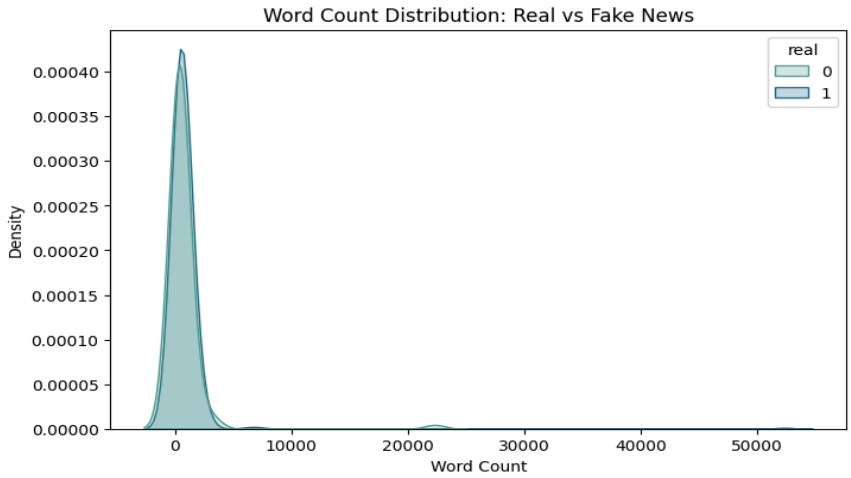
Tweet Count by News Authenticity



Note. This is the distributed social media interactions of the news in the dataset separated according to the feature real to be true or false (real or fake)

Figure 4

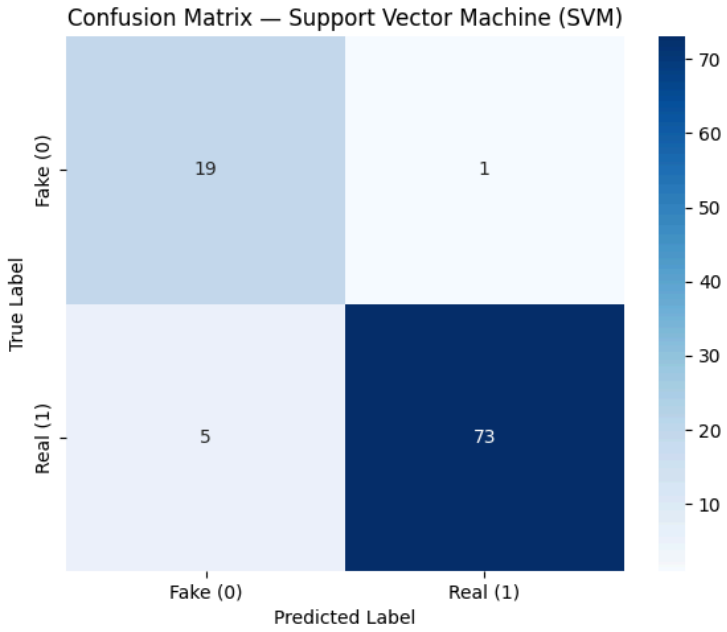
Word Count Distribution: Real vs Fake News



Note. This is the word count distribution of the news in the dataset separated according to the feature real to be true or false (real or fake)

Figure 5

Confusion Matrix — Support Vector Machine (SVM)



Note. This is how the best model algorithm (SVM) performed in the test phase which can be used to calculate precision, recall, accuracy and the F1 score

Discussion and Limitations

The findings of this research directly answer the research question that was determined. Using social media features (tweet_num) and textual analysis at the same time helps the model to be able to forecast the fake news automatically. The highest F1 score achieved by the proposed model was 96.05% while the highest F1 score of those existing models in the literature was 74.60% by exBAKE and 87.00% to 88.00% by TriFN (It should be noted that the existing models in the literature were evaluated on general news articles while the proposed model was evaluated on a dataset that contains only financial news). These results suggest that social media spread may be a consequential feature in this process since the proposed model can find stronger signals compared to the existing models in the literature which does not use the spread amount feature. However, no experiment was conducted that directly compared the performance of a textual analysis model with a model that combines the textual analysis with the spread amount. Therefore, this relationship cannot be proven. In an economical perspective, fake financial news being filtered helps individual investors to have a more trustworthy, stable and fair environment in the markets by reducing the panic causing changes. The limitations that were faced in this research were:

1. To be able to filter financial news in a practical way, some financial keywords were searched. However there may be some other news articles about finance that do not contain these words, so there may be more data that could be used but deleted. This leads to the loss of some data, which means the model could not learn with its full capacity.
2. The social media metrics are built up in a time so for the real-world usage of the model it may need some time for the model to be able to make more accurate predictions for the fake news detections. This leads the model not to be able to be used directly when news articles are just published.

3. The textual analysis also provides a strong signal for the model to be able to guess the reality state of the news article; however it does not mathematically or socially prove that there is a logical correlation. This means that when some patterns change the model's accuracy could be crucially affected.

Ethical Discussion

There may be some bias of the model to think that a news article is real against the option fake because there are more real news data in the dataset that was used in this research compared to fake news. However, the dataset consists of real-world data which means in the real world there is also less fake news than real news since people can recognize some of them and they cannot get spread as much as the real ones. One of the ethical problems this model can create is letting people be able to understand how the fake news is detected by AI models and create fake news articles which are less likely to be detectable. There are more false positives (FP) than the false negatives in the final performance of the model. When real news articles are detected as fake this situation is FP. When fake news articles are not detected as fake, this situation is FN. There is a chance of 6.41% that the model will label a legit news article as fake (FP) and 5% chance that it will label a fake news article as real (FN). The FP thwarts the true information flow in the market creating fake panic leading to artificial and unexpected price movements. The FN can result in the perception of manipulative news as trustable, which can cause investors to make wrong decisions and lose money.

Conclusion

The main purpose of this research project is to predict the truthfulness of news articles by analyzing various characteristics of the news, such as its spread and the text of the article. The project aims to develop novel AI models that have not yet been explored in the existing literature. According to the Efficient Market Hypothesis, the market is shaped by information. Fake news creates artificial volatilities in the market which means they change the market artificially in a way that should not happen

normally. This artificiality manipulates individual investors and stock predicting models to analyze the market incorrectly resulting in an artificial market. The real world aim of this project is creating a fair, trustworthy and transparent investing environment for everyone.

The hybrid model developed in this study demonstrates that financial fake news can be detected with high accuracy by combining pre-trained BERT text representation models with social media interaction metrics (tweet counts) and textual features (word count). Experimental results show that the Linear Kernel Support Vector Machine (SVM) model exhibits significantly superior performance compared to the K-Nearest Neighbor (KNN) model, with 93.88% accuracy and a 96.05% F1 score. This reveals that the SVM architecture, supported by balanced class weights to counter data imbalance, is highly effective in capturing fake news patterns in high-dimensional text vector spaces.

The findings indicate that the speed of dissemination and text characteristics on social media can be used as powerful signals in preventing financial misinformation. Automatically filtering misleading financial news contributes to reducing artificial volatility, protecting individual investors from manipulation, and creating a fairer and more transparent trading environment in financial markets. In future studies, incorporating the temporal dynamics of social media interactions (time series analysis) into the model and testing different Deep Learning-based natural language processing architectures on larger, financially focused datasets could further enhance the model's predictive power in real-time market conditions.

References

Alghamdi, J., Luo, S., & Lin, Y. (2024). A comprehensive survey on machine learning approaches for fake news detection. *Multimedia Tools and Applications*, 83(17), 51009-51067.

algord. (2020). *Fake news dataset*. Kaggle. https://www.kaggle.com/datasets/algord/fake-news

Chang, Y. T., Hu, Z., Li, X., Yang, S., Jiang, J., & Sun, N. (2024, October). Dihan: A novel dynamic hierarchical graph attention network for fake news detection. In Proceedings of the 33rd ACM International Conference on Information and Knowledge Management (pp. 197-206).

Jwa, H., Oh, D., Park, K., Kang, J. M., & Lim, H. (2019). exBAKE: Automatic fake news detection model based on bidirectional encoder representations from transformers (bert). *Applied Sciences*, 9(19), 4062.

Rusli, N. I. A., Abdullah, N. A. S., Abd Razak, F. N., & Ramli, N. H. (2025). Machine learning methods for detecting fake news: A systematic literature review of machine learning applications in key domains. *Machine Learning*, 16(7).

Shu, K., Sliva, A., Wang, S., Tang, J., & Liu, H. (2017). Fake news detection on social media: A data mining perspective. *ACM SIGKDD Explorations Newsletter*, 19(1), 22-36.

Shu, K., Wang, S., & Liu, H. (2019, January). Beyond news contents: The role of social context for fake news detection. In Proceedings of the twelfth ACM international conference on web search and data mining (pp. 312-320).

Titan, A. G. (2015). The efficient market hypothesis: Review of specialized literature and empirical research. *Procedia Economics and Finance*, 32, 442-449.